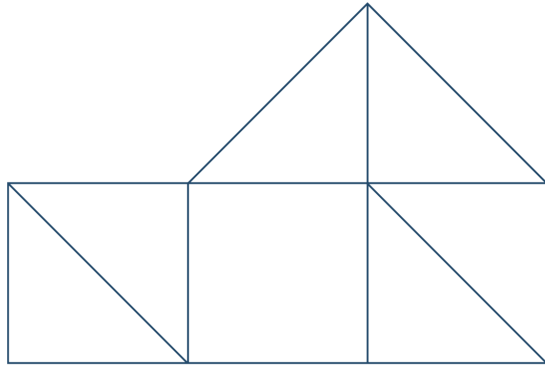


Exercise Sheet #2: DC 1: Circuit Calculations

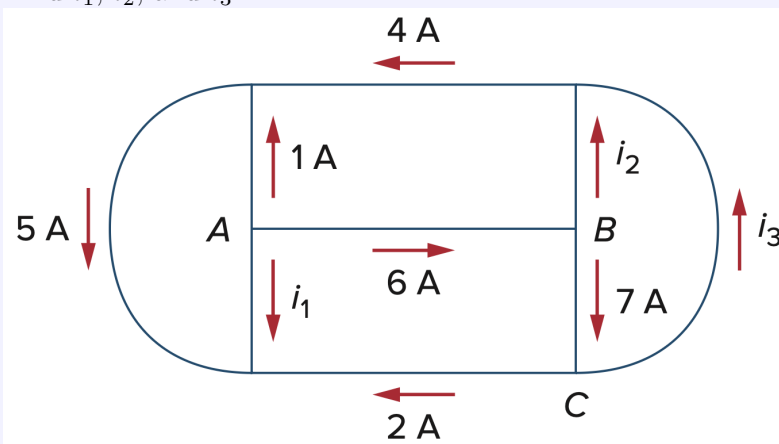
Discussion: Oct 22, 2025

Exercise 1: Topology

Find the number of nodes, branches, and loops.

**Solution**

$$n = 9, b = 15, l = 7$$

Exercise 2: KCLFind i_1 , i_2 , and i_3 .

Solution

$$-1\text{ A} - 6\text{ A} - i_1 = 0$$

$$\Rightarrow i_1 = -1\text{ A} - 6\text{ A} = -7\text{ A}$$

$$7\text{ A} - 2\text{ A} = 5\text{ A} - i_3 = 0$$

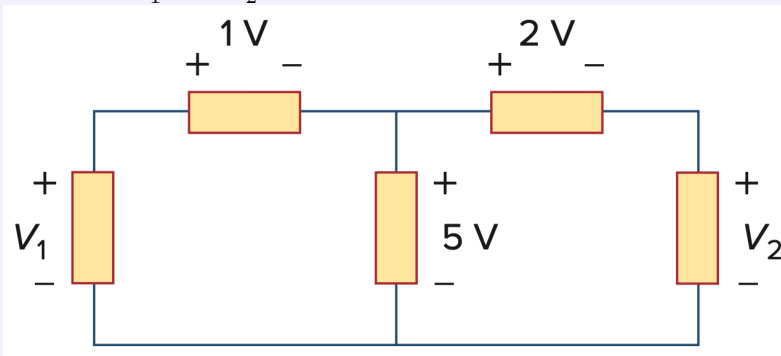
$$\Rightarrow i_3 = 7\text{ A} - 2\text{ A} = 5\text{ A}$$

$$i_2 + i_3 - 4\text{ A} = 0$$

$$\Rightarrow i_2 = 4\text{ A} - i_3 = -1\text{ A}$$

Exercise 3: KVL

Calculate V_1 and V_2 .

**Solution**

$$1\text{ V} + 5\text{ V} - V_1 = 0$$

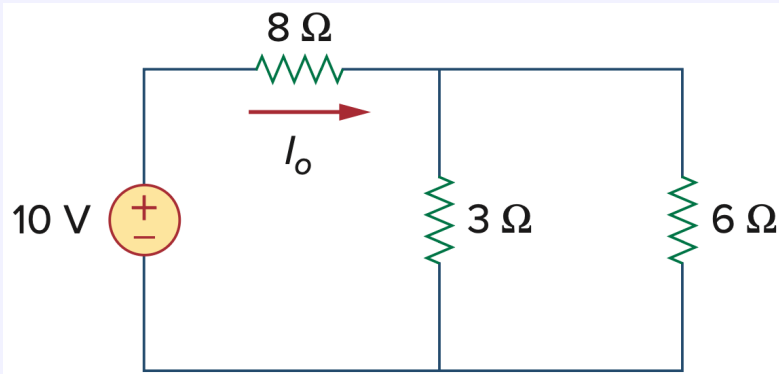
$$\Rightarrow V_1 = 1\text{ V} + 5\text{ V} = 6\text{ V}$$

$$2\text{ V} + V_2 - 5\text{ V} = 0$$

$$\Rightarrow V_2 = 5\text{ V} - 2\text{ V} = 3\text{ V}$$

Exercise 4: Series and Parallel Resistors

Calculate I_0 .



Solution

First, we calculate the equivalent resistance of the parallel resistors with 3Ω and 6Ω , then we add the series resistor with 8Ω :

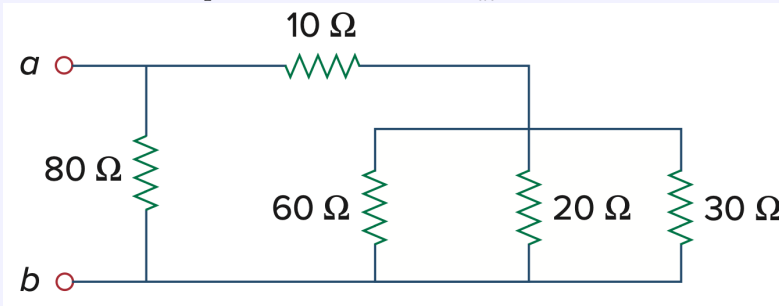
$$R = \frac{3\Omega \cdot 6\Omega}{3\Omega + 6\Omega} + 8\Omega = 10\Omega$$

Finally, we can calculate the current I_o using Ohm's law:

$$I_o = \frac{10\text{ V}}{10\Omega} = 1\text{ A}$$

Exercise 5: Equivalent Resistance

Calculate the equivalent resistance R_{ab} across the terminals.



Solution

First we calculate the equivalent resistance of the parallel resistors with 60Ω , 20Ω , and 30Ω :

$$R_p = \frac{1}{\frac{1}{60\Omega} + \frac{1}{20\Omega} + \frac{1}{30\Omega}} = 10\Omega$$

Then we add the series resistor with 10Ω :

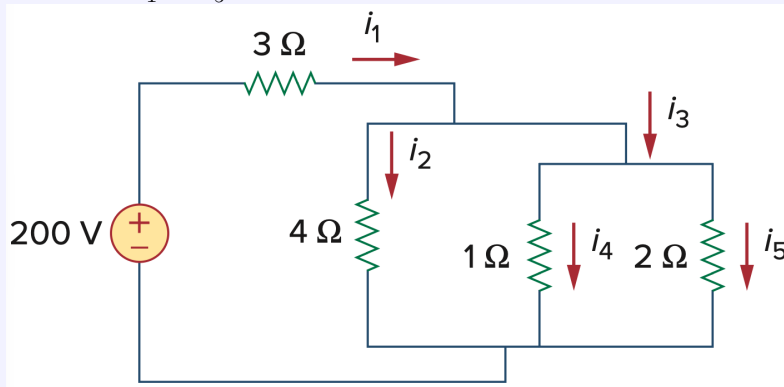
$$R_{ab} = 10\Omega + 10\Omega = 20\Omega$$

And finally we combine with the parallel resistor of 80Ω :

$$R_{ab} = \frac{1}{\frac{1}{20\Omega} + \frac{1}{80\Omega}} = 16\Omega$$

Exercise 6: Current Divider Rule

Calculate i_1 to i_5 .

**Solution**

First, we calculate the equivalent resistance of the parallel resistors with $1\ \Omega$ and $2\ \Omega$ and call it R_3 :

$$R_3 = \frac{1\ \Omega \cdot 2\ \Omega}{1\ \Omega + 2\ \Omega} = \frac{2}{3}\ \Omega$$

Then we calculate the total resistance R from the $4\ \Omega$ resistor parallel to R_3 and the $3\ \Omega$ resistor in series:

$$R = 3\ \Omega + \frac{4\ \Omega \cdot \frac{2}{3}\ \Omega}{4\ \Omega + \frac{2}{3}\ \Omega} = \frac{25}{7}\ \Omega$$

Using Ohm's law, we can calculate the total current i_1 :

$$i_1 = \frac{200\ \text{V}}{\frac{25}{7}\ \Omega} = 56\ \text{A}$$

Then, we can calculate i_2 using current division:

$$i_2 = i_1 \cdot \frac{R_3}{R_2 + R_3} = 56\ \text{A} \cdot \frac{\frac{2}{3}\ \Omega}{4\ \Omega + \frac{2}{3}\ \Omega} = 8\ \text{A}$$

KCL gives us i_3 :

$$i_3 = i_1 - i_2 = 56\ \text{A} - 8\ \text{A} = 48\ \text{A}$$

Again, we can calculate i_4 using current division:

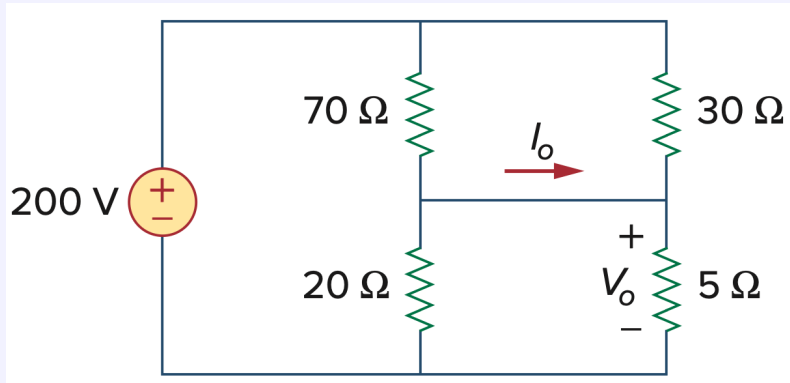
$$i_4 = i_3 \cdot \frac{R_5}{R_4 + R_5} = 48\ \text{A} \cdot \frac{2\ \Omega}{1\ \Omega + 2\ \Omega} = 32\ \text{A}$$

Finally, KCL gives us i_5 :

$$i_5 = i_3 - i_4 = 48\ \text{A} - 32\ \text{A} = 16\ \text{A}$$

Exercise 7: Node Connection

Calculate I_o and V_o .



Solution

A node is a point where two or more circuit elements connect (electrically the same point). A branch is a path that connects two nodes and contains a circuit element (resistor, source, etc.).

Here, there are three main nodes:

- Top node: connected to the +200 V source, top of the 70 Ω and 30 Ω resistors.
- Bottom node: 0 V reference.
- Middle node: connection between two branches. On the left, the branch formed by the 70 Ω and 20 Ω . And on the right, the branch formed by the 30 Ω and 5 Ω .

The middle horizontal wire (with I_o) is part of the middle node, but because there's a potential difference between the left and right parts of that node before it's connected, current flows through that connection — hence we analyze I_o as if it's a branch current.

As the output voltage V_o is the voltage across the 5 Ω resistor, the output voltage equals the middle node voltage.

Currents from the top node to the middle node:

$$I_{70} = \frac{200 \text{ V} - V_o}{70 \Omega}, \quad I_{30} = \frac{200 \text{ V} - V_o}{30 \Omega}$$

Currents from the middle node to the bottom node:

$$I_{20} = \frac{V_o}{20 \Omega}, \quad I_5 = \frac{V_o}{5 \Omega}$$

Applying KCL at the middle node:

$$\frac{200 \text{ V} - V_o}{70 \Omega} + \frac{200 \text{ V} - V_o}{30 \Omega} = \frac{V_o}{20 \Omega} + \frac{V_o}{5 \Omega}$$

Simplify:

$$\frac{200 \text{ V} - V_o}{21 \Omega} = \frac{V_o}{4 \Omega}$$

$$\Rightarrow 800 \text{ V} - 4 \cdot V_o = 21 \cdot V_o$$

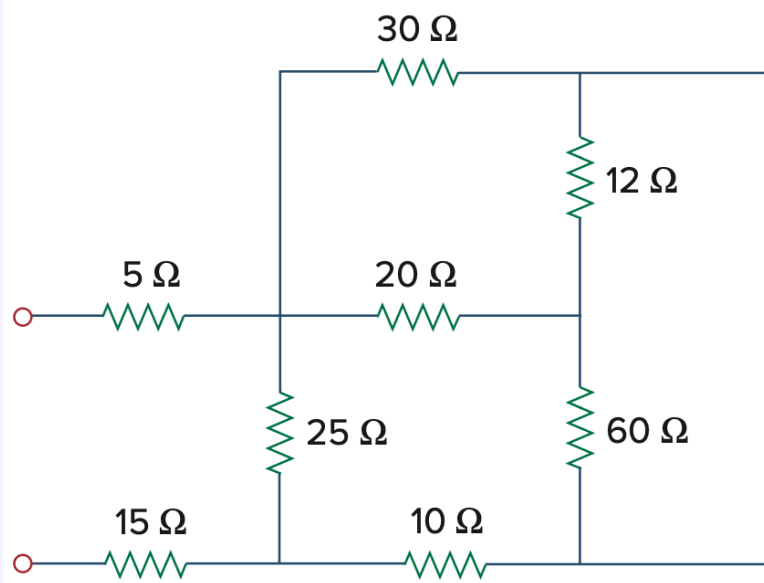
$$\Rightarrow \boxed{V_o = 32 \text{ V}}$$

Now we use KCL to compute the current I_o (defined to the right):

$$I_o = I_{70} - I_{20} = \frac{200 \text{ V} - 32 \text{ V}}{70 \Omega} - \frac{32 \text{ V}}{20 \Omega} = \boxed{0.8 \text{ A}}$$

Exercise 8: Δ -Y Transformation

Calculate the equivalent resistance across the terminals.



Solution

1. Combine the right vertical resistors:

$$12\,\Omega \parallel 60\,\Omega = \frac{12\,\Omega \cdot 60\,\Omega}{12\,\Omega + 60\,\Omega} = 10\,\Omega$$

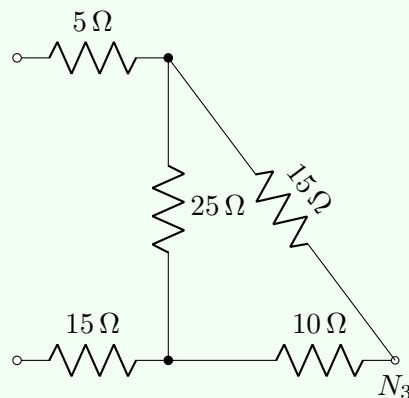
2. Series combination with the $20\,\Omega$ resistor:

$$20\,\Omega + 10\,\Omega = 30\,\Omega$$

3. Parallel combination with the other $30\,\Omega$:

$$30\,\Omega \parallel 30\,\Omega = \frac{30\,\Omega \cdot 30\,\Omega}{30\,\Omega + 30\,\Omega} = 15\,\Omega$$

After these three steps, the circuit reduces to a triangle (Δ):

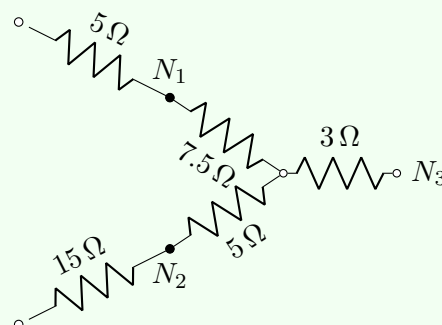


4. Convert the Δ -network to equivalent Y-resistances:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{15\,\Omega \cdot 25\,\Omega}{50\,\Omega} = 7.5\,\Omega$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} = \frac{10\,\Omega \cdot 25\,\Omega}{50\,\Omega} = 5\,\Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{10\,\Omega \cdot 15\,\Omega}{50\,\Omega} = 3\,\Omega$$



The branch with the $3\,\Omega$ resistor connects to a dead-end node N_3 , so it carries no current and can be ignored.

$$\Rightarrow R_{eq} = 5\,\Omega + 7.5\,\Omega + 5\,\Omega + 15\,\Omega = 32.5\,\Omega$$